Olympiad Test Paper — Science (Class 7) — Paper 5

Chapter 2: Nutrition in Animals

Paper 5 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	2 — Nutrition in Animals
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. These are the hardest traps: deciding which single step a multi-event scenario isolates and proving the earlier steps still ran; separating WHERE enzymes sit (gut cavity vs cells vs blood) from WHAT they do; telling physical emulsification from chemical breakdown; sorting egestion from excretion when shed cells or fibre are involved; untangling rumen/abomasum and cud/chyme; reasoning quantitatively about teeth and surface area; and mapping Amoeba's single-cell, temporary-structure plan onto a human's many fixed organs. Do not write on this paper — mark answers on your answer sheet.

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1. A radioactive tag is placed on one molecule of butter in a slice of toast and tracked at four moments: (i) the toast is chewed and swallowed, (ii) an enzyme splits the butter molecule into smaller units, (iii) those units pass through the intestinal wall into a transport fluid, (iv) a cell stitches them into its own outer boundary. Name the four steps (i)–(iv), in order.

(A) Ingestion, digestion, assimilation, absorption
(B) Digestion, ingestion, absorption, egestion
(C) Ingestion, absorption, digestion, assimilation
(D) Ingestion, digestion, absorption, assimilation

2. Tests on a weakening patient show: appetite and meals normal; faeces contain no unused nutrients; blood sugar stays high and steady around the clock; yet the muscles keep shrinking. Which single step has failed, and which finding rules OUT an absorption failure?

(A) Absorption has failed, shown by the high blood sugar

(B) Assimilation has failed; the high, steady blood sugar proves nutrients ARE crossing into the blood (absorption works), so only the cells' use of them is broken

(C) Digestion has failed, shown by the shrinking muscles

(D) Egestion has failed, shown by the normal faeces

3. Four things the body gets rid of: (i) plant fibre passed in faeces, (ii) carbon dioxide breathed out, (iii) dead gut-lining cells passed in faeces, (iv) nitrogen waste filtered from the blood. Sort them correctly into egestion and excretion.

(A) Egestion: (i),(ii); Excretion: (iii),(iv)

(B) Egestion: (i),(iii) — matter leaving the food canal that was never taken into and used by the body's cells; Excretion: (ii),(iv) — wastes made by the body's own cells

(C) All four are egestion, since all leave the body

(D) All four are excretion, since all are unwanted

4. A leech sucks blood, a sea anemone sweeps tiny prey into its mouth with stinging tentacles, and a sparrow swallows a whole seed. A judge says one nutrition step is shared by all three at the moment shown, despite their utterly different tools. Which step, and why do the different tools not matter?

(A) Digestion — each chemically breaks the food as it is taken

(B) Ingestion — each is simply food being taken into the body; sucking, sweeping and swallowing are only different methods of the same taking-in step

(C) Absorption — each passes nutrients into body fluids at the feeding point

(D) Assimilation — each puts the food to use as it is gathered

5. An inventor imagines an animal that squirts enzymes onto food lying OUTSIDE its body, lets it break down in a puddle, and claims the nutrients then soak straight from the puddle into its blood, with no gut at all. Using the five steps, why can this not nourish the animal?

(A) Absorption means nutrients crossing the gut wall INTO the body; an outside puddle has no body wall to cross, so the broken-down food must still be taken in (ingestion) before any absorption — outside soaking cannot feed it

(B) It works, because digesting the food outside is enough to nourish the animal

(C) It fails only because enzymes cannot act outside a body

(D) It fails because food must be assimilated before it is digested

6. A human uses legs to approach food, hands to grasp it, and a stomach to digest it. An Amoeba performs the first two of these with ONE structure it forms only when needed. What body-plan point does this make, and which of the three jobs does that structure NOT perform?

(A) The structure digests and absorbs at once, doing two chemical jobs

(B) The structure is a permanent miniature organ used only for grasping

(C) The structure makes the Amoeba's own bile, so it can digest fat without a liver

(D) Being one cell, the Amoeba forms a single temporary structure (pseudopodium) for BOTH moving and grasping instead of a fixed organ per task; it does not digest — that happens later inside a food vacuole

7. Besides tasting, the tongue is essential for swallowing a dry biscuit. A student lists only 'it tastes the food.' Which TWO further jobs is the tongue doing, and why is swallowing impossible without them?

(A) It makes the saliva itself and absorbs the biscuit's sugar into the blood

(B) It mixes the biscuit with saliva and rolls it into a soft ball, then pushes that ball to the back of the mouth to be swallowed; without mixing and pushing, the dry crumbs cannot be gathered or sent down

(C) It digests the starch into sugar and then cools the biscuit

(D) It produces acid to soften the biscuit and pushes air to swallow

8. Three facts about a tooth: its outer layer resists a steel file the longest; the layer beneath is softer and forms most of the tooth's bulk; the innermost space bleeds and hurts when reached. Name the layers from outside in, AND say which fact pins down the pulp.

(A) Dentine, enamel, pulp; the 'resists a file' fact pins the pulp

(B) Enamel, pulp, dentine; the 'softer bulk' fact pins the pulp

(C) Pulp, dentine, enamel; the 'bleeds and hurts' fact pins the enamel

(D) Enamel (hardest, outer), dentine (softer bulk), pulp (innermost); the 'bleeds and hurts when reached' fact pins the pulp, since the nerves and vessels lie there

9. A tooth is X-rayed: decay has tunnelled almost to the centre on one side but the patient feels nothing; on the other side a hairline crack barely dents the surface yet triggers sharp pain when cold water touches it. A student says the deep side must be made of a tougher material. What really explains both observations?

(A) The deep side is tougher dentine, so it resists all sensation

(B) The painful crack reached nerves that are stored in the enamel

(C) Both cannot be true; deeper decay must always hurt more

(D) Pain comes only from reaching the nerves in the central pulp: the deep tunnel has stopped just short of the pulp, while the hairline crack has reached or nearly reached it

10. Decay runs: sugar left on teeth → bacteria eat it → bacteria make acid → acid eats enamel. A school can fund only ONE measure. Rank these by how early in the chain they act, and pick the earliest-acting one: (i) fluoride paste that hardens enamel, (ii) a rule to rinse the mouth right after lunch, (iii) gum that 'tires' the bacteria so they make less acid.

(A) (i) acts earliest, by strengthening the enamel before any acid forms

(B) (ii) acts earliest, because rinsing removes the leftover sugar — the very first link — before bacteria can even feed

(C) (iii) acts earliest, by stopping the bacteria from making acid

(D) None acts early; only repairing dissolved enamel helps

11. A museum labels two jaws only by tooth shape. Jaw 1: long blade-like teeth at the front sides, with few flat teeth. Jaw 2: wide ridged grinding teeth filling the back, with blunt small front teeth. From teeth alone, which jaw ate flesh and which ate plants?

(A) Jaw 1 ate flesh (blade-like side teeth tear meat) and Jaw 2 ate plants (wide ridged teeth grind); tooth form tracks diet

(B) Jaw 2 ate flesh, since only meat-eaters have many teeth

(C) Both jaws ate the same food; tooth shape is mere decoration

(D) Nothing about diet can be told from teeth alone

12. A person bolts a chapati in two big lumps without chewing; another chews the same chapati thoroughly. Both have identical saliva and stomachs, yet the chewer's starch is digested faster. Give the precise reason, in terms of what chewing does for the enzyme.

(A) Chewing adds extra amylase released from the teeth

(B) Chewing makes the chapati acidic, which speeds the enzyme

(C) Chewing breaks the chapati into many small pieces, exposing far more surface for the salivary (and later) enzymes to act on, so breakdown is faster

(D) Chewing warms the chapati, and the heat alone digests the starch

13. A student tests three foods by chewing each a long time: a plain rice ball turns sweet, a boiled egg-white does not, a spoon of ghee does not. What single conclusion about saliva's enzyme do all three results together support?

(A) Saliva digests every food, so the egg and ghee secretly sweetened too

(B) Saliva acts mainly on protein, which is why the egg should have sweetened

(C) Saliva acts on fat, so the ghee should have tasted sweetest

(D) Saliva's enzyme acts only on starch (sweetening the rice) and not on protein (egg) or fat (ghee), which is why only the rice turned sweet

14. Three warm test tubes of starch: Tube 1 gets fresh saliva, Tube 2 gets saliva first boiled then cooled, Tube 3 gets only warm water. After thirty minutes only Tube 1 shows sugar. Which single conclusion is fully supported, and what is Tube 3's job?

(A) Warmth made the sugar, since all three tubes were warm

(B) Fresh saliva's enzyme made the sugar (boiling destroyed it in Tube 2), and Tube 3 is the control proving warm water alone does nothing

(C) Saliva's acid, not an enzyme, made the sugar in Tube 1

(D) Boiled saliva is stronger, so Tube 2 should have shown the most sugar

15. Two volunteers swallow a dry cream-cracker. Volunteer 1 has normal saliva and swallows easily; Volunteer 2's salivary glands are temporarily blocked and the dry crumbs catch painfully in the throat. Besides starting starch digestion, what saliva job does this reveal, and why does swallowing depend on it?

(A) Saliva moistens and binds the dry crumbs into a slippery soft lump that slides down the food pipe; without it (Volunteer 2) the dry crumbs cannot be swallowed smoothly

(B) Saliva absorbs the cracker's sugar into the blood right in the mouth

(C) Saliva makes hydrochloric acid in the mouth to soften the cracker

(D) Saliva cools the cracker so it cannot scratch the throat

16. A cow grazes with its head DOWN and its stomach ABOVE its mouth, yet swallowed grass still travels up to the stomach. A student says gravity helps because the grass is heavy. Which single point most decisively refutes this?

(A) With the head down, gravity pulls the grass back toward the mouth, AWAY from the higher stomach; so the grass reaching the stomach must be pushed by the food pipe's muscular waves, not pulled by gravity

(B) Gravity still helps, since heavy grass always slides toward the stomach

(C) The cow's food pipe is a straight vertical drop, so its position is irrelevant

(D) The food pipe absorbs the grass directly, so it never travels to the stomach

17. A drug paralyses ONLY the food-pipe muscles (no more squeezing waves); the mouth, stomach, intestines, saliva and all juices keep working. A boy chews and swallows a piece of banana. What is the most direct problem, and what proves the stomach itself is fine?

(A) The stomach stops churning, so the banana ferments inside the food pipe

(B) The swallowed banana cannot be pushed down the food pipe and lodges there; that the stomach still churns whatever reaches it proves the fault is transport in the pipe, not the stomach

(C) Salivary digestion stops, turning the banana bitter in the mouth

(D) Water is over-absorbed in the food pipe, drying the banana into a hard lump

18. A swallowed dye is timed: it flashes past one organ in a few seconds, lingers in the next for about three hours, and stays in the third for the better part of a day. Name the three organs from shortest to longest stay, and which is the dye's longest residence.

(A) Stomach (seconds), oesophagus (hours), small intestine (most of a day)

(B) Oesophagus (seconds), stomach (a few hours), small intestine (most of a day — its longest residence)

(C) Oesophagus (seconds), small intestine (a few hours), stomach (most of a day)

(D) Small intestine (seconds), stomach (hours), oesophagus (most of a day)

19. A chart lists gastric-juice parts and a claimed job for each. HCl: 'kills swallowed germs.' Mucus: 'digests the meat's protein.' Pepsin: 'begins breaking down protein.' HCl (again): 'keeps the stomach acidic for pepsin.' Exactly one job is wrong. Which, and what is that part's real job?

(A) 'Mucus digests the meat's protein' is wrong; mucus coats and protects the wall, while pepsin is the protein-digesting enzyme

(B) 'HCl kills swallowed germs' is wrong

(C) 'Pepsin begins breaking down protein' is wrong

(D) 'HCl keeps the stomach acidic for pepsin' is wrong

20. A medicine strongly suppresses stomach acid for months. Predict the TWO most direct effects, each traced to a real job of the acid.

(A) Starch digestion fails everywhere, and bile can no longer reach the intestine

(B) Fat can no longer be emulsified, and large-intestine water absorption stops

(C) Stomach protein digestion weakens (pepsin needs the acid) and swallowed germs are less effectively killed

(D) Saliva stops being made, and the food pipe loses its peristalsis

21. An empty stomach, between meals, still contains acid and pepsin, yet it does not digest its own protein wall. A student says 'an empty stomach makes no pepsin, so there is nothing to attack the wall.' Why is that wrong, and what truly protects the wall?

(A) The student is right; an empty stomach has no pepsin

(B) Bile from the liver flows up and neutralises the acid between meals

(C) Even between meals acid and pepsin are present; the wall IS protein but is shielded by a constant mucus layer lining its inner surface

(D) The wall is made of bone, which pepsin cannot digest

22. Pepsin works only in strong acid; the small intestine's enzymes work only in non-acidic surroundings. As the half-digested protein paste leaves the stomach for the small intestine, what MUST change for digestion to continue, and what would happen to the intestinal enzymes if it did not?

(A) It must become even more acidic; otherwise the intestinal enzymes stop

(B) Its strong acid must be neutralised in the small intestine; if it stayed acidic, the intestinal enzymes (which need non-acid conditions) could not work

(C) It must stay exactly as acidic as the stomach so the same enzymes continue

(D) It must be dried out completely before the intestinal enzymes can act

23. Match three terms to three descriptions: (i) the paste a human's stomach sends on to the intestine; (ii) the grass a cow brings back up to re-chew; (iii) the fat-emulsifying fluid from the liver. Choose the row that labels all three correctly.

(A) (i) cud, (ii) chyme, (iii) bile
(B) (i) bile, (ii) cud, (iii) chyme
(C) (i) cud, (ii) bile, (iii) chyme
(D) (i) chyme, (ii) cud, (iii) bile

24. Two drainpipes of equal length carry equal water for equal time: Pipe A is smooth inside; Pipe B's inner wall is lined with thousands of tiny folds and finger-like bumps. Pipe B lets far more water seep through its wall. Mapping this to the gut, the single best reason villi help absorption is that they:

(A) Greatly increase the inner surface area, so far more nutrient-absorbing contact fits into the same length of gut

(B) Speed the food through, so it wastes less time in the gut

(C) Make the gut more acidic, which dissolves nutrients faster

(D) Produce extra bile to break down the fats

25. Two patients eat identical, well-digested meals. Patient 1 has healthy villi; Patient 2's villi are flattened smooth. Blood tests show Patient 2's nutrient levels rising far less after meals, and over months Patient 2 loses weight while Patient 1 holds steady. What does this contrast pin down as the villi's role?

(A) The villi digest the food; without them digestion fails

(B) The villi churn the food; without them the stomach must work harder

(C) The villi provide the surface that ABSORBS digested nutrients into the blood; flatten them and absorption — and weight — fall even though digestion is normal

(D) The villi make bile; without them fat is not absorbed

26. After a meal, two events are recorded an hour apart: first, digested amino acids pass from the gut into the bloodstream; later, a healing wound knits new skin using those amino acids. A student calls both 'absorption.' Give the correct labels for the two events.

(A) Both are absorption, since amino acids move in both

(B) The first (amino acids entering the blood) is absorption; the second (skin cells using them to build new tissue) is assimilation

(C) Both are assimilation, since the body benefits in both

(D) The first is digestion and the second is absorption

27. A student reasons: 'Simple sugars and amino acids are first detected at the villi, so the villi must be where digestion finishes.' Correct the reasoning, and name where the digestion that produced those molecules actually happened.

(A) The molecules are first detected at the villi only because that is where they are ABSORBED; the digestion that produced them happened earlier, in the cavity of the gut, before the food reached the villi

(B) The student is right; the villi are the body's main digesting organs

(C) The villi both digest and store the molecules, which is why they appear there

(D) The villi churn the food like a small stomach to finish digestion

28. In the small intestine, compare two happenings: (i) a large fat drop is broken by bile into a mist of tiny droplets, the fat molecules themselves still whole; (ii) a glucose molecule passes through the wall into the blood. A student calls both 'digestion.' Label each correctly.

(A) Both are absorption, since both move material in the intestine

(B) Both are digestion, since both change the food in the intestine

(C) (i) is part of digestion — a physical preparation of the fat for enzymes (the molecules are not yet cut); (ii) is absorption — the glucose crossing into the blood — not digestion

(D) (i) is absorption and (ii) is digestion

29. A student argues: 'The small intestine is several times longer than the large intestine, and longer organs handle the bigger jobs, so the small intestine must be where most water is reclaimed too.' Pinpoint the error.

(A) The student is right; the longer small intestine reclaims the water as well

(B) The error is only about 'bigger jobs'; the small intestine still reclaims most of the water

(C) Length does not assign jobs: the small intestine specialises in absorbing nutrients, while reclaiming water from the residue is specifically the large intestine's job

(D) Both reclaim water equally, so neither is specially responsible

30. In a gut model, the large-intestine section's water-reclaiming pump can be turned up or down. Predict the stool when the pump reclaims far too LITTLE water, and contrast it with reclaiming far too MUCH.

(A) Too little gives hard, dry stool; too much gives watery stool

(B) Too little causes acid vomiting; too much speeds fat digestion

(C) Either way the stool is unchanged; the large intestine ignores water

(D) Too little reclaimed water leaves loose, watery stool; too much reclaimed water leaves hard, dry stool (constipation)

31. A person eats a salad full of plant fibre (cellulose), which human enzymes cannot break. Following the five steps, what happens to the fibre, and what useful role does it play on the way despite never being digested?

(A) It is digested by stomach acid and absorbed as sugar

(B) It is assimilated into the body's cells for energy

(C) It is excreted as a chemical waste made inside the body's cells

(D) It is ingested and finally egested undigested (no human enzyme breaks cellulose); on the way it adds bulk that helps move the residue along the gut, but it is never absorbed or assimilated

32. The liver is the body's largest gland and makes bile, yet bile carries NO digestive enzyme. A student concludes bile does nothing for digestion. Refute this and state exactly what bile does and where.

(A) Bile, made by the liver and acting in the small intestine, physically breaks large fat globules into tiny droplets, giving fat-digesting enzymes far more surface to work on — so it helps greatly without being an enzyme

(B) The student is right; with no enzyme, bile is useless

(C) Bile chemically cuts fat into amino acids in the stomach

(D) Bile turns fat into starch so that starch enzymes can finish it

33. A patient's bile-storing sac is removed. Over a year, only large oily meals cause trouble (loose, greasy stools); rice and dal meals are handled normally. What does this selective pattern reveal about what the sac stored and what that fluid chiefly helps digest?

(A) The sac stored bile, and bile chiefly helps digest fats — so only oily meals are noticeably affected

(B) The sac made pepsin, a fat-digesting enzyme, so fatty meals suffer

(C) The sac stored amylase, so starchy meals should have suffered instead

(D) The sac produced stomach acid, so all three meal types should suffer equally

34. A meal of roti (starch), dal (protein) and a little oil (fat) is fully broken down. In which single organ is the digestion of ALL THREE finally COMPLETED, just before absorption?

(A) The small intestine, where pancreatic and intestinal juices (with bile for the fat) finish all three

(B) The stomach, where acid and pepsin finish all three

(C) The mouth, where saliva finishes all three

(D) The large intestine, as the water is absorbed

35. A student writes: 'All three of saliva, pepsin and bile chemically cut their food into smaller molecules.' Exactly one of the three does NOT cut molecules chemically. Which one, and what does it do instead?

(A) Saliva does not cut chemically; it only wets the food

(B) Pepsin does not cut chemically; it only churns the protein

(C) All three cut chemically, so the student is right

(D) Bile does not cut molecules chemically; it only physically emulsifies fat into tiny droplets, leaving the fat molecules whole

36. A doctor explains why the failure of ONE gland can hurt the digestion of starch, protein AND fat all at once, while losing the saliva would hurt only the starch. Which gland is being described, and why does its failure hit all three foods?

(A) The salivary gland, because it makes the enzymes for all three foods

(B) The pancreas: its juice supplies the small intestine with enzymes for carbohydrates, proteins and fats, so its failure impairs all three together — unlike saliva, which acts only on starch

(C) The liver, because its bile chemically digests all three foods

(D) The stomach, because its pepsin digests all three foods

37. Trace where digestive juices are ADDED to food along the canal. Three are added in order: (1) an enzyme for starch, (2) acid plus a protein enzyme, (3) bile plus a juice acting on all three foods. Name the organ that adds each, in the order 1 → 2 → 3.

(A) Stomach, then mouth, then small intestine

(B) Small intestine, then stomach, then mouth

(C) Mouth (saliva, the starch enzyme), then stomach (acid + pepsin), then small intestine (bile + the pancreatic/intestinal juices for all three)

(D) Mouth, then small intestine, then stomach

38. A student labels a cow's stomach chambers: 'cellulose is fermented by microbes in the abomasum, and acid-and-enzyme digestion happens in the rumen.' Both labels are swapped. State the correct chamber for each job.

(A) The labels are correct as given

(B) Only the first is wrong: the microbes work in the reticulum

(C) Only the second is wrong: the acid digestion is in the omasum

(D) Both are swapped: microbes ferment cellulose in the RUMEN (the largest chamber), and acid-and-enzyme digestion happens in the ABOMASUM

39. A buffalo and a child are fed only fresh grass for a month. The buffalo stays strong; the child grows weak and malnourished. A student says 'the buffalo just has a bigger stomach.' Give the deeper, correct reason.

(A) The buffalo has rumen microbes plus cud re-chewing that together break down grass's cellulose; the child has neither and cannot digest cellulose, so gains little from grass

(B) The buffalo's stomach is merely larger, so it holds more grass

(C) The child has no stomach in which to hold the grass

(D) Grass needs no digestion; the buffalo simply eats more of it

40. A goat brings grass back up and chews it very finely for many minutes. A student concludes the long mouth-chewing is what digests the cellulose. Yet measurements show most cellulose breakdown happens in the largest stomach chamber, not the mouth. Correct the student's view.

(A) The student is right; the long chewing digests the cellulose in the mouth

(B) The cellulose is broken down in the food pipe during re-swallowing

(C) Mouth-chewing only grinds the grass finer; the cellulose itself is broken down by microbes in the rumen, not in the mouth

(D) The cellulose is never digested; the goat just stores the cud forever

41. Re-chewing the cud is said to let a cow free MORE energy from grass. Lay out the cause-and-effect chain by which finer grinding ends in more energy released.

(A) Finer grinding turns the cellulose straight into sugar in the mouth, absorbed there

(B) Finer grinding exposes more grass surface → rumen microbes and enzymes reach and break down more of the cellulose → more usable nutrients are freed → more energy is released

(C) Finer grinding heats the grass, and the heat releases the energy

(D) Finer grinding turns the cellulose into protein, which holds more energy

42. An Amoeba pushes its undigested leftovers out when its food vacuole meets the cell surface; a human passes undigested residue out at the end of the gut. A student says these are different steps because one happens in a single cell and one in an organ. Which step are BOTH performing, and what is the Amoeba's pocket called?

(A) Both perform excretion; the pocket is a gall bladder

(B) Both perform absorption; the pocket is a villus

(C) Both perform EGESTION (removing undigested leftovers); in the Amoeba the leftovers were held in a food vacuole

(D) Both perform assimilation; the pocket is a rumen

43. In an Amoeba, digestive enzymes are poured INTO the food vacuole — a closed pocket WITHIN the cell but sealed off from the cell's own working parts. The closest human parallel for WHERE digestive enzymes act is:

(A) Inside the blood, acting on nutrients after they are absorbed

(B) Inside the body's muscle cells, where nutrients are burnt for energy

(C) On the outer skin surface, before the food is swallowed

(D) Inside the cavity of the stomach and intestine — enclosed spaces that, like the vacuole, are walled off from the body's own working cells and from the blood

44. A debater claims: 'An Amoeba, though just one cell, carries out every one of the five nutrition steps a human does — the only difference is the machinery.' Pick the statement that BOTH backs the claim and states the difference correctly.

(A) Only the human truly digests; the Amoeba just engulfs and stores food

(B) The Amoeba absorbs but never egests, so it completes only four of the five steps

(C) Both ingest, digest, absorb, assimilate and egest; the Amoeba does all five inside one cell, while the human spreads them across many specialised organs

(D) The human never assimilates, so neither completes all five steps

45. A thali holds rice (starch), paneer (protein) and ghee (fat). For each food, name the organ where its chemical digestion FIRST begins. Which row is correct?

(A) Rice → mouth; paneer → stomach; ghee → small intestine

(B) Rice → stomach; paneer → mouth; ghee → stomach

(C) Rice → small intestine; paneer → small intestine; ghee → mouth

(D) All three → the stomach

46. A patient's stomach makes no pepsin, but the small intestine's protein enzymes work normally. For a protein meal, what happens to protein digestion, and where is it nonetheless completed and absorbed?

(A) Protein is not digested at all, since pepsin is essential everywhere

(B) Protein digestion does not BEGIN in the stomach (no pepsin), but the small intestine's enzymes can still complete it, and the building blocks are absorbed there

(C) Protein digestion shifts entirely to the mouth instead

(D) Protein is absorbed undigested in the large intestine

47. Of the dissolved glucose and the bulk water in a juicy meal, one is taken up mostly high in the gut and the other mostly low in the gut. A student says 'water is the bigger thing, so it must be absorbed first, before the small glucose.' Correct this with the right sites.

(A) Glucose is mainly absorbed earlier, in the small intestine; most remaining water is reclaimed later, in the large intestine — so the student's order is backwards

(B) The student is right; water is absorbed before the glucose

(C) Both are absorbed together in the stomach(D) Both are absorbed in the mouth

48. A spoonful of dal has just entered the mouth and chewing has begun. A student says 'two steps are done — I've ingested it and saliva has digested it.' Exactly how many of the five steps are COMPLETED, and what is the precise flaw in the student's claim?

(A) Two are complete, exactly as the student says

(B) One step (ingestion) is complete, and digestion is finished too

(C) Zero steps are COMPLETED: ingestion is still happening, and although saliva has STARTED on the starch, that digestion is not finished — so no step is yet complete

(D) Four steps are complete and only egestion remains

49. Two students reorder the five steps. Student P writes: ingestion, digestion, assimilation, absorption, egestion. Student Q writes: ingestion, absorption, digestion, assimilation, egestion. Each made exactly one impossible swap. Which pair of fixes is correct?

(A) P is fine; Q must move egestion earlier

(B) Both must put ingestion last instead of first

(C) P swapped absorption and assimilation (a cell cannot use a nutrient before it enters the blood); Q swapped digestion and absorption (a molecule cannot cross the gut wall before it is broken small)

(D) Neither student made any error

50. A student labels three glands: 'the pancreas makes bile, the liver stores it, and the gall bladder's juice digests all three food types.' Every one of the three is wrong. Give the correct gland for each role, in the order: bile-maker, bile-storer, all-three-juice gland.

(A) Pancreas, gall bladder, liver

(B) Liver (makes bile), gall bladder (stores bile), pancreas (juice for all three)

(C) Gall bladder, liver, pancreas (D) Liver, pancreas, gall bladder

51. After a meal the stomach's muscular wall squeezes and churns for hours. A student says 'this churning is the same as the food pipe's swallowing waves — both just move food along.' What is the churning's distinct purpose that the swallowing waves do NOT serve?

(A) Churning only moves food along, exactly like the swallowing waves

(B) Churning absorbs nutrients into the blood, unlike swallowing

(C) Churning mixes the food thoroughly with gastric juice (acid and pepsin) and breaks it into a fine paste — mixing-and-mechanical-breakdown that the food pipe's simple forward waves do not do

(D) Churning makes bile, which the swallowing waves do not

52. A frog flicks out a sticky tongue to catch a fly, while an Amoeba flows around a particle to engulf it. A student says these two differ in their step of DIGESTION. Pinpoint the conceptual mistake.

(A) They differ in their method of INGESTION (how food is taken in), not digestion; digestion is the later chemical breakdown, which these clips do not show differing

(B) They differ in absorption, since each moves nutrients into its fluids differently

(C) The student is right; tongue-catching and engulfing are two kinds of digestion

(D) They differ in assimilation, since each builds tissue from food in its own way

53. A person eats well and their gut digests and absorbs normally, but their cells cannot put the absorbed glucose to use, so they weaken. In a DIFFERENT person, the gut absorbs nothing into the blood at all. Which step is failing in each person, respectively?

(A) In the first person absorption is failing; in the second, digestion is failing

(B) In both people the same step, digestion, is failing

(C) In the first person assimilation is failing; in the second, absorption is failing

(D) In the first person egestion is failing; in the second, ingestion is failing

54. A student claims 'animals make their own food inside their bodies from sunlight, just more slowly than plants do.' Correct this with the right term for how animals obtain food and what it means.

(A) Correct; animals photosynthesise slowly inside their cells

(B) Wrong; animals are heterotrophs — they cannot make their own food and must take in ready-made food (from plants or other animals), unlike plants which make theirs

(C) Wrong; animals make their food in the stomach from water alone

(D) Correct; the liver makes food from sunlight

55. A child loses the use of every canine but keeps all other teeth. A flesh-eating diet would suffer because canines do what, and which kept tooth type cannot substitute for that exact job?

(A) Canines grind plant food; the incisors cut and so cannot grind in their place

(B) Canines cut food at the front; the molars tear and so cannot cut in their place

(C) Canines absorb nutrients in the mouth; no other tooth can absorb in their place

(D) Canines tear flesh; the flat-topped molars and premolars grind rather than tear, so they cannot substitute for the tearing

56. A poster shows food meeting the organs in this order: mouth → stomach → large intestine → small intestine → out. Exactly one pair of organs is in the wrong order. Which correction is needed?

(A) The mouth and stomach should swap(B) Nothing is wrong; the order is correct

(C) The stomach should come after the large intestine

(D) The small intestine and large intestine are swapped; food reaches the small intestine BEFORE the large intestine, so the order is mouth → stomach → small intestine → large intestine → out

57. An adult human has 32 teeth made of incisors, canines, premolars and molars. If exactly one quarter of all the teeth are the pointed-and-front cutting-and-tearing kinds combined, how many teeth are grinders, and does this match the standard set?

(A) 8 are cutting-and-tearing and 24 are grinders, which exactly matches the standard set

(B) 24 are cutting-and-tearing and 8 are grinders, matching the standard set

(C) 8 teeth are cutting-and-tearing, so 24 would be grinders by this quarter-rule, which does NOT match the standard set of 12 cutting-and-tearing and 20 grinders

(D) 16 are cutting-and-tearing and 16 are grinders, matching the standard set

58. In an Amoeba the digesting (in the food vacuole) and the using of nutrients both happen in the SAME single cell. In a human, a biologist says, the two are kept in separate places. Which pairing of human locations shows this separation?

(A) Digestion inside the body cells; the nutrients used in the gut cavity

(B) Digestion in the blood; the nutrients used in the gut cavity

(C) Both digestion and nutrient use happen only in the mouth

(D) Digestion in the cavity of the gut; the nutrients later used inside the body's own cells, reached through the blood

59. A textbook groups a butterfly's sipping tube, an Amoeba's pseudopodia, and a cow's broad grinding teeth under one heading: 'Different tools for the same step, shaped by the food eaten.' Name the step, and explain why grinding teeth belong beside a sipping tube and pseudopodia.

(A) Digestion — each tool chemically breaks the food as it handles it

(B) Absorption — each tool passes nutrients into body fluids at the feeding point

(C) Assimilation — each tool lets the body use the food as it is gathered

(D) Ingestion — all are tools for taking food IN; the cow's grinding teeth are part of how it takes food in, by preparing the grass for swallowing

60. A revision table has three cells to fill: 'protein digestion begins in ___', 'fat digestion is mainly completed in ___', and 'water is chiefly recovered in ___'. A student fills them as mouth / stomach / mouth, and gets all three wrong. Give the correct three, in order.

(A) Stomach; small intestine; large intestine(B) Mouth; stomach; small intestine

(C) Small intestine; mouth; large intestine(D) Large intestine; stomach; mouth

Solutions — Science (Class 7), Chapter 2: Nutrition in Animals — Paper 5

Paper 5 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	A	21	C	31	D	41	B	51	C
2	B	12	C	22	B	32	A	42	C	52	A
3	B	13	D	23	D	33	A	43	D	53	C
4	B	14	B	24	A	34	A	44	C	54	B
5	A	15	A	25	C	35	D	45	A	55	D
6	D	16	A	26	B	36	B	46	B	56	D
7	B	17	B	27	A	37	C	47	A	57	C
8	D	18	B	28	C	38	D	48	C	58	D
9	D	19	A	29	C	39	A	49	C	59	D
10	B	20	C	30	D	40	C	50	B	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) Chewing and swallowing the toast is ingestion; the enzyme cutting the butter molecule into smaller units is digestion; those units crossing the intestinal wall into the transport fluid is absorption; a cell stitching them into its own boundary is assimilation — so the order is ingestion, digestion, absorption, assimilation. The list that puts assimilation before absorption is impossible, because a cell cannot use the units until they have first entered the fluid that reaches it. Putting digestion before ingestion is wrong, since nothing can be broken down before it is taken in. Putting absorption before digestion fails, since a large molecule cannot cross the wall until it has been cut small.

2. (B) The high, steady blood sugar is the key clue: its presence in the blood proves the food was taken in, broken down, and carried across the gut wall, so ingestion, digestion and absorption all worked. The wasting muscles despite that sugar show the cells are not putting it to use — the assimilation step. Absorption cannot be the failure, precisely because the sugar DID reach the blood. Digestion cannot be the failure, since the sugar (a digested product) is present. Egestion is unrelated, since the faeces are normal and the problem is with sugar already in the blood.

3. (B) Egestion removes matter that passed through the food canal without ever being taken into and used by the body's living cells; that fits the plant fibre (i) and the shed gut-lining cells (iii), which simply travelled out in the residue. Excretion removes wastes the body's own cells produced by their chemistry; that fits the carbon dioxide (ii) and the nitrogen waste (iv). Grouping (ii) with egestion is wrong, since CO_2 is made inside cells. Calling all four egestion ignores that CO_2 and nitrogen waste are cell-made; calling all four excretion ignores that fibre and shed cells were never used inside cells.

4. (B) All three clips show food being taken into the body, which is ingestion; sucking (leech), sweeping with tentacles (anemone) and swallowing whole (sparrow) are just different methods of the same taking-in step. They are not digestion, since no chemical breakdown is shown at the moment of feeding. They are not absorption, since nutrients are not crossing into body fluids at the feeding point. They are not assimilation, since no cell is yet using anything.

5. (A) Absorption, by definition, is nutrients crossing the gut wall INTO the body; an outside puddle offers no body wall to cross, so nothing can be absorbed there. The broken-down food in the puddle must still be taken in (ingestion) before it can be absorbed inside — so external soaking cannot nourish the animal. Digesting food outside is not enough on its own. Enzymes can in fact act outside a body (as a housefly shows), so that is not the reason. And food can never be assimilated before it is digested, so that ordering is impossible too.

6. (D) The point is about body plan: being a single cell, the Amoeba forms one temporary structure — the pseudopodium — that does double duty for both moving toward food and grasping it, instead of keeping a separate fixed organ for each task as a many-organ body does. That structure does NOT digest; digestion happens later inside a food vacuole. So the 'digests and absorbs at once' claim is wrong, the 'permanent miniature organ' claim is wrong (its whole point is that it is temporary), and the 'makes its own bile' claim is wrong, since an Amoeba has no liver and the bile idea does not apply to a single cell.

7. (B) Beyond tasting, the tongue mixes the dry biscuit with saliva and rolls it into a soft, slippery ball, then pushes that ball to the back of the mouth so it can be swallowed; without the mixing and the pushing, the dry crumbs could neither be gathered nor sent down. The tongue does not make the saliva (the salivary glands do) nor absorb sugar into the blood in the mouth. It does not itself digest the starch (saliva's enzyme does) or cool the food. And it makes no acid; that belongs to the stomach.

8. (D) The outer layer that resists a file longest is enamel, the hardest substance in the body; beneath it lies the softer dentine that forms most of the tooth's bulk; the innermost space that bleeds and hurts when reached is the pulp, because the nerves and blood vessels lie there — so 'bleeds and hurts when reached' is exactly what pins down the pulp. Starting with dentine outside enamel is wrong, since dentine lies under the enamel. Putting the pulp before the dentine is impossible, since the pulp is innermost. And the fully reversed order puts the soft inner pulp outside the hard enamel, which cannot be.

9. (D) Pain in a tooth comes only from reaching the nerves in the central pulp, not from how deep or wide the decay looks. The deep tunnel that hurts nothing has stopped just short of the pulp; the hairline crack that hurts has reached or nearly reached it (cold water on the exposed pulp triggers the pain). 'Tougher dentine resists sensation' is false and irrelevant — enamel, not dentine, is the hardest, and hardness is not what causes pain. The nerves are in the pulp, not the enamel. And depth alone does not decide pain, so both observations can indeed be true at once.

10. (B) The chain starts with sugar left on the teeth, so removing that sugar by rinsing (ii) breaks the very first link, before the bacteria can even feed — the earliest possible action. Hardening enamel with fluoride (i) acts only at the last link, after acid has formed. 'Tiring' the bacteria with gum (iii) acts in the middle, after the bacteria already have the sugar. So ranked by earliness, rinsing comes first, and it is the single measure to fund.

11. (A) Long blade-like side teeth are suited to tearing flesh, so Jaw 1 fits a flesh-eater; wide ridged grinding teeth at the back suit grinding plant matter, so Jaw 2 fits a plant-eater — tooth form tracks diet. The number of teeth says nothing about whether an animal eats meat, so 'only meat-eaters have many teeth' is unwarranted. The two jaws are described with clearly different teeth, so they did not eat the same food. And tooth shape is not decoration; it reliably reflects diet.

12. (C) Chewing breaks the chapati into many small pieces, which exposes far more surface for the salivary enzyme (and later enzymes) to act on, so the starch is broken down faster — this is why the chewer digests it sooner. Teeth do not release amylase, so 'extra amylase from the teeth' is false. Chewing does not make food acidic. And while chewing may warm the food slightly, heat alone does not digest starch — an enzyme does. The real link is mechanical breakdown increasing surface area for the enzyme.

13. (D) Only the starchy rice ball turned sweet, because saliva's enzyme breaks starch into sugar; the egg-white (protein) and the ghee (fat) have nothing that enzyme can act on, so neither sweetened. Saliva does not digest every food, so the egg and ghee would not secretly sweeten. The enzyme does not act on protein, so the egg would not sweeten by that route. And it does not act on fat, so the ghee cannot taste sweetest. The three results together pin the enzyme to starch alone.

14. (B) All three tubes were warm, yet only the one with fresh saliva made sugar, so warmth is not the cause (ruling out A). Tube 2's saliva had its enzyme destroyed by boiling, so the difference between Tubes 1 and 2 shows the enzyme is what made the sugar. Tube 3 (warm water only) is the control: it shows that warm water by itself produces no sugar, isolating saliva as the active agent. Acid is not the cause, since boiling removes the enzyme but not acid, yet Tube 2 still failed. And boiled saliva is not stronger; it has lost its active enzyme.

15. (A) The contrast between the two volunteers reveals saliva's second job: it moistens and binds the dry crumbs into a slippery, soft lump that slides easily down the food pipe. Volunteer 2's blocked glands leave the crumbs dry, so they catch in the throat — proving swallowing depends on that moistening. Saliva does not absorb the cracker's sugar into the blood in the mouth (absorption happens far downstream). It does not make hydrochloric acid (that is the stomach's), and cooling is not a recognised swallowing role.

16. (A) With the cow's head down and stomach above the mouth, gravity pulls the swallowed grass back toward the mouth, AWAY from the higher stomach; so the grass that still reaches the stomach must be pushed there by the food pipe's muscular waves (peristalsis), not pulled by gravity. Saying gravity helps ignores that gravity is working against the grass here. The food pipe is not a straight vertical drop, and even if it were, the inverted position clearly matters. And the food pipe does not absorb the grass into the blood; the grass genuinely travels to the stomach.

17. (B) The food pipe moves food by peristaltic muscle waves; paralyse those and the swallowed banana cannot be pushed down, so it lodges in the pipe — the most direct problem. That the stomach still churns whatever reaches it shows the stomach is working normally, so the fault is transport in the pipe, not the stomach. Salivary digestion still works (the mouth is unaffected), so the banana does not turn bitter for that reason. And the food pipe is not where water is absorbed, so over-drying there is not the mechanism.

18. (B) The wave-pushing tube the dye flashes through in seconds is the oesophagus; the organ it lingers in for a few hours is the stomach; the one it stays in for most of a day, for digestion and absorption, is the small intestine — so shortest to longest is oesophagus, stomach, small intestine, and the small intestine is the longest residence. The stomach is not a seconds-long passage, and the small intestine (not the stomach) is the long-stay organ, so the orders that swap them are wrong.

19. (A) Mucus does not digest anything; its job is to coat and shield the stomach wall, so 'mucus digests the meat's protein' is the wrong assignment — pepsin is the actual protein enzyme. 'HCl kills swallowed germs' is correct. 'Pepsin begins breaking down protein' is correct. 'HCl keeps the stomach acidic for pepsin' is correct. Only the mucus statement is false.

20. (C) The acid has two jobs: it makes the acidic conditions pepsin needs, and it kills germs in food. Suppress it and pepsin works poorly, so stomach protein digestion weakens, and swallowed germs are less effectively killed — the two most direct effects. Starch digestion does not depend on stomach acid, and bile's path to the intestine is unrelated to it. Fat emulsification depends on bile, not acid, and water absorption is the large intestine's separate job. Saliva and peristalsis are not produced by stomach acid.

21. (C) Even between meals the stomach holds acid and pepsin, so 'an empty stomach makes no pepsin' is false. The wall really is made of protein; what saves it is the constant mucus layer lining the inner surface, which keeps the acid and pepsin from reaching the living tissue. Bile is made in the liver and acts in the small intestine, not as a stomach coating. And the wall is not bone; it is protein tissue protected by mucus.

22. (B) The small intestine's enzymes work only in non-acid surroundings, so the strongly acidic paste leaving the stomach must have its acid neutralised once it reaches the small intestine; if it stayed acidic, those intestinal enzymes could not work and digestion would stall. Making it more acidic would suit only pepsin, not the intestinal enzymes. Keeping it stomach-acidic would block the intestinal enzymes. And drying it out is not how digestion proceeds; the paste stays moist.

23. (D) The paste a human stomach sends on to the intestine is chyme; the grass a cow brings back up to re-chew is cud; the fat-emulsifying fluid from the liver is bile — so the correct row is (i) chyme, (ii) cud, (iii) bile. The rows that call the human paste 'cud' or 'bile' mislabel it; cud is specifically the ruminant's regurgitated grass, and bile is the liver's fluid, not a stomach paste. Only the row matching chyme, cud and bile to (i), (ii) and (iii) is right.

24. (A) The folds and finger-like bumps lining Pipe B multiply the inner surface area, so far more of the wall is in contact with the contents and far more can seep through in the same length — exactly how villi aid absorption in the gut. They do not speed the food through (both pipes share equal time), do not change acidity (absorption is not a matter of dissolving in acid), and do not make bile (bile comes from the liver). The single best reason is the increased surface area.

25. (C) The two-patient contrast isolates the villi's role: with the same well-digested meals, the only difference is the villi, and the patient who lost them absorbs far less and loses weight, while the other holds steady. So the villi provide the surface that ABSORBS digested nutrients into the blood; flatten them and absorption — and weight — fall even though digestion is normal. The villi do not digest the food (digestion is normal in both), do not churn it (that is the stomach), and do not make bile (the liver does). The decisive role is absorption surface.

26. (B) Amino acids passing from the gut into the blood is absorption — nutrients crossing the gut wall. The skin cells later using those amino acids to build new tissue is assimilation — the body putting absorbed nutrients to use. So the two events are absorption then assimilation, not both 'absorption.' They are not both assimilation, since the first is merely entering the blood, not yet being used. And the first is not digestion (the amino acids are already digested products), so 'digestion then absorption' is wrong.

27. (A) The simple molecules are first detected at the villi only because that is where they are ABSORBED into the blood; the digestion that produced them happened earlier, by enzymes acting in the cavity of the gut, before the food reached the villi. So the villi absorb, they do not digest. They are not the body's main digesting organs (digestion is in the cavity), they do not store the molecules (these pass into the blood), and they do not churn food like a stomach.

28. (C) Breaking a fat drop into tiny droplets is a physical preparation within digestion — the fat molecules are not yet cut, only spread out for enzymes to reach — so (i) is part of digestion. A glucose molecule crossing the wall into the blood is absorption, by definition, not digestion — so (ii) is absorption. Both are not absorption (emulsifying fat crosses no wall). Both are not digestion (crossing into blood is absorption). And the labels are not reversed, since emulsifying stays in the cavity while crossing the wall is absorption.

29. (C) Organ length does not decide which job an organ does. The small intestine specialises in absorbing nutrients, while reclaiming water from the residue is specifically the large intestine's job — so the longer small intestine does NOT also reclaim the water. The error is not merely about 'bigger jobs'; it is a real division of labour. And the two do not reclaim water equally; it is chiefly the large intestine's task.

30. (D) Water left in the residue makes the stool loose and watery, so reclaiming too LITTLE water gives watery stool; reclaiming too MUCH dries the residue into hard stool (constipation). The reversed pairing is wrong, since more water reclaimed means a drier — not wetter — stool. Acid vomiting and fat digestion are unrelated to large-intestine water recovery. And the large intestine certainly does handle water; reclaiming it is its main job.

31. (D) No human enzyme can break cellulose, so the salad fibre is taken in (ingestion) and finally passed out undigested with the residue (egestion); it is never digested, absorbed or assimilated. On the way, though, it adds bulk that helps push the residue along the gut — a useful role even though it yields no nutrients. Stomach acid does not turn fibre into sugar; the body cannot assimilate undigested fibre for energy; and it is not excretion, since excretion removes waste made inside cells, not undigested food passing through the canal.

32. (A) Bile helps without being an enzyme: made by the liver and acting in the small intestine, it physically breaks large fat globules into tiny droplets, vastly increasing the surface on which fat-digesting enzymes can work — so it is far from useless. The student is therefore wrong. Bile does not chemically cut fat into amino acids (amino acids come from protein, and bile carries no enzyme), and it does not convert fat into starch, which is not a real process.

33. (A) Because only large oily meals are troublesome (greasy stools) while rice and dal are fine, the lost fluid must be one that mainly helps fat: that fluid is bile, which the sac stored, and bile assists fat digestion by emulsifying it. The sac did not make pepsin (a stomach protein enzyme, not a fat enzyme). It did not store amylase; if it had, starchy meals would suffer, which they do not. And it did not make stomach acid; the selective effect on fats rules out an effect on all foods.

34. (A) The digestion of all three foods is finally COMPLETED in the small intestine, where the pancreatic and intestinal juices (with bile preparing the fat) finish carbohydrate, protein and fat digestion just before absorption. The stomach only churns and starts protein digestion, not all three. The mouth only starts starch digestion. And the large intestine recovers water; it does not complete digestion. The small intestine is the common finishing site.

35. (D) Saliva's enzyme chemically cuts starch, and pepsin chemically cuts protein — both break molecules into smaller ones. Bile is the odd one out: it does NOT cut molecules chemically; it only physically emulsifies fat into tiny droplets, leaving the fat molecules whole. So 'saliva only wets food' is false (it has an enzyme), 'pepsin only churns protein' is false (it cuts protein), and 'all three cut chemically' is false because bile does not. Bile is the one that acts physically.

36. (B) The gland is the pancreas: its juice supplies the small intestine with enzymes for carbohydrates, proteins and fats, so when it fails, all three are impaired together — whereas saliva acts only on starch, so losing it would hurt only the starch. The salivary gland does not make enzymes for all three, so it is not the answer. Bile (from the liver) emulsifies fat physically and carries no enzyme that chemically digests all three. And the stomach's pepsin digests only protein, not all three. Only the pancreas fits the "hits all three at once" clue.

37. (C) Juices are added in this order along the canal: first the mouth adds saliva with the starch enzyme; then the stomach adds acid and pepsin; then the small intestine receives bile and the pancreatic/intestinal juices that act on all three foods. So the order of organs is mouth, stomach, small intestine. The other orders misplace the stomach or the small intestine ahead of the mouth, which cannot be, since food meets the mouth first and the small intestine last of these three.

38. (D) Both labels are swapped: microbes ferment the cellulose in the RUMEN, the largest chamber, and acid-and-enzyme digestion like a human stomach happens in the ABOMASUM. So 'correct as given' is wrong, and the microbes are not in the reticulum, nor is the acid digestion in the omasum. The only fully correct fix names the rumen for the microbial cellulose breakdown and the abomasum for the acid-and-enzyme digestion.

39. (A) Grass is rich in cellulose, which neither animal's own enzymes break; the buffalo manages only because microbes in its rumen, plus repeated cud-chewing, break the cellulose down — a system the child lacks entirely. A merely bigger stomach would not help, since the limit is cellulose-digesting microbes, not capacity. The child does have a stomach. And grass certainly needs digestion; eating more of an indigestible food does not nourish.

40. (C) Chewing the cud only grinds the grass finer; the actual breakdown of cellulose is done by microbes in the rumen (the largest chamber), which is exactly why measurements show most breakdown there, not in the mouth. The mouth has no cellulose-digesting power. The food pipe merely transports the cud. And the cellulose is not stored unchanged forever; it is broken down in the rumen, helped by the finer grinding.

41. (B) The chain runs: grinding the cud finer exposes more grass surface → rumen microbes and enzymes can reach and break down more of the cellulose → more usable nutrients are freed → more energy is released. The mouth does not turn cellulose into sugar for absorption there; that breakdown happens in the rumen. Chewing does not release energy by heat. And finer grinding does not convert cellulose into protein; that conversion does not occur.

42. (C) Both the Amoeba and the human are performing EGESTION — removing undigested leftovers from the body. The fact that one happens in a single cell and the other in an organ does not change the step; only the body plan differs. In the Amoeba, the leftovers were held in a food vacuole, which is what the pocket is called. The pocket is not a gall bladder, villus or rumen, and the step is not excretion (cell-made waste), absorption (taking in) or assimilation (using nutrients).

43. (D) The food vacuole is an enclosed pocket where enzymes meet food, walled off from the cell's own working parts; the closest human parallel is the enclosed cavity of the stomach and intestine, which is likewise walled off from the body's working cells and from the blood. Human digestive enzymes do not act in the blood (which carries already-absorbed nutrients), nor inside muscle cells (where nutrients are merely burnt), nor on the skin surface (food is digested inside the gut).

44. (C) Both the Amoeba and the human carry out all five steps — ingest, digest, absorb, assimilate and egest; the only difference is that the Amoeba does them inside one cell while the human spreads them across many specialised organs, which is exactly the claim. The Amoeba does digest (in its food vacuole), so 'never digests' is false; it does egest leftovers, so it completes all five (not four); and the human does assimilate, using absorbed nutrients in its cells, so 'never assimilates' is false.

45. (A) Starch (rice) first begins digesting in the mouth with the salivary enzyme; protein (paneer) first begins in the stomach with pepsin; fat (ghee) is first chemically digested in the small intestine (after bile emulsifies it). So the correct row is rice → mouth, paneer → stomach, ghee → small intestine. The other rows misplace the starting organs — for instance putting protein in the mouth or fat in the stomach — which do not match where each food's digestion actually begins.

46. (B) Pepsin normally starts protein digestion in the stomach; with no pepsin, that beginning step is lost. But the small intestine's protein enzymes work normally, so they can still complete the protein's digestion, and the building blocks are absorbed there at the villi. So protein is not left wholly undigested (pepsin is not essential everywhere), digestion does not shift to the mouth (saliva acts on starch, not protein), and protein is not absorbed undigested in the large intestine.

47. (A) Dissolved glucose is mainly absorbed high in the gut, across the villi of the small intestine; most of the remaining water is reclaimed lower down, in the large intestine — so the student's 'water first' order is backwards. Being a 'bigger thing' does not decide the order, and water is not a big molecule anyway. Both are not absorbed in the stomach, and neither is mainly absorbed in the mouth. Glucose comes first (small intestine), water later (large intestine).

48. (C) At the instant the dal enters the mouth and chewing has just begun, ingestion is still happening — it is not yet complete — and although saliva has STARTED breaking the starch, that digestion is far from finished, so no step has been completed. The student's flaw is treating 'started' as 'completed': beginning to ingest and beginning to digest do not make two finished steps. So zero steps are complete, not two; not one with digestion finished; and certainly not four.

49. (C) Student P put assimilation before absorption, which is impossible: a cell cannot use a nutrient until it has crossed into the blood (absorption must precede assimilation). Student Q put absorption before digestion, which is impossible: a molecule cannot cross the gut wall until it has been broken small (digestion must precede absorption). So each made one impossible swap, and the correct fixes name exactly those two. P is not 'fine,' neither order needs ingestion last, and both clearly contain an error.

50. (B) Every one of the student's labels is wrong. The largest gland that MAKES bile is the liver; the small sac that STORES bile is the gall bladder; the gland whose juice acts on all three food types is the pancreas. So in the order bile-maker, bile-storer, all-three-juice gland, the answer is liver, gall bladder, pancreas. The other orders keep one or more of the student's mix-ups — for instance making the pancreas the bile-maker or the gall bladder the all-foods gland — which do not match the real roles.

51. (C) The stomach's churning is not just transport: it mixes the food thoroughly with gastric juice (acid and pepsin) and mechanically breaks it into a fine paste. The food pipe's swallowing waves only push food forward; they do not mix or grind it. So churning is not 'exactly like' the swallowing waves, it does not absorb nutrients into the blood (the small intestine does that), and it does not make bile (the liver does). Its distinct purpose is the mixing-and-mechanical-breakdown.

52. (A) Catching a fly with a sticky tongue and engulfing a particle with pseudopodia are two ways of taking food in, so the frog and the Amoeba differ in their method of

INGESTION, not digestion. Digestion is the later chemical breakdown, which neither clip shows differing. They do not differ in absorption (nutrients crossing into body fluids, not yet shown) and they are not two kinds of digestion. Assimilation — cells using nutrients — is also not what these feeding clips depict.

53. (C) In the first person, glucose is absorbed into the blood but the cells cannot use it, so the failing step is assimilation; in the second, nothing crosses into the blood at all, so the failing step is absorption. The first is not absorption, because the glucose did reach the blood. Neither failure is digestion, since food is broken down in both descriptions. And egestion and ingestion are not the failing steps, since the problems lie with using glucose and with crossing into the blood.

54. (B) Animals are heterotrophs: they cannot make their own food and must take in ready-made food, whether from plants or from other animals — unlike plants, which make their own food. So the student is wrong; animals do not photosynthesise, slowly or otherwise. They do not make food in the stomach from water, and the liver does not make food from sunlight. The correct correction names heterotrophic nutrition and what it means.

55. (D) Canines are the pointed teeth that tear flesh; the molars and premolars are flat grinders, so they cannot take over the tearing job — which is why a flesh-eating diet would suffer when the canines are lost. Canines do not grind (that is the molars' role), so the grinding claim is wrong. Canines tear rather than cut; cutting is the incisors' job. And teeth do not absorb nutrients at all, so the absorption claim is false.

56. (D) The poster swapped the small intestine and large intestine: food reaches the small intestine BEFORE the large intestine, so the correct order is mouth → stomach → small intestine → large intestine → out. The mouth and stomach are already in the right order, so swapping them is wrong. The order is not correct as shown, so "nothing is wrong" fails. And the stomach already comes before both intestines, so moving it after the large intestine is wrong. Only the small-intestine/large-intestine swap needs fixing.

57. (C) One quarter of 32 is 8, so the quarter-rule would give 8 cutting-and-tearing teeth and 24 grinders; but the real adult set has 8 incisors plus 4 canines, that is 12 cutting-and-tearing, and 20 grinders — so the quarter-rule does NOT match. Saying 24 grinders matches is wrong, since the true count is 20. Reversing to 24 cutting teeth is far from the real 12. And a 16-16 split matches neither the rule nor the real set.

58. (D) In a human the two are kept apart: digestion takes place in the enclosed cavity of the gut, while the nutrients are later used inside the body's own cells, reached through the blood — unlike the Amoeba, where both happen in one cell. So digestion is not inside the body cells, nor in the blood, and using the nutrients is not done in the gut cavity, which rules out the reversed pairings; and neither digestion nor nutrient use is confined to the mouth.

59. (D) A sipping tube, pseudopodia and grinding teeth are all tools for taking food IN, so the shared step is ingestion; the cow's grinding teeth belong beside the others because grinding the grass is part of how the cow takes its food in, preparing it for swallowing. They are not tools of digestion (chewing prepares food, but the chemical breakdown comes later), not of absorption (nothing crosses into body fluids at the mouth), and not of assimilation (which is cells using nutrients).

60. (A) Protein digestion begins in the stomach (with pepsin), fat digestion is mainly completed in the small intestine (with bile and enzymes), and water is chiefly recovered in the large intestine — so the correct three are stomach, small intestine, large intestine. Protein does not begin in the mouth (saliva acts on starch), fat is not completed in the stomach or mouth, and water is recovered in the large intestine, not the mouth — so the student's mouth / stomach / mouth answers are all wrong.